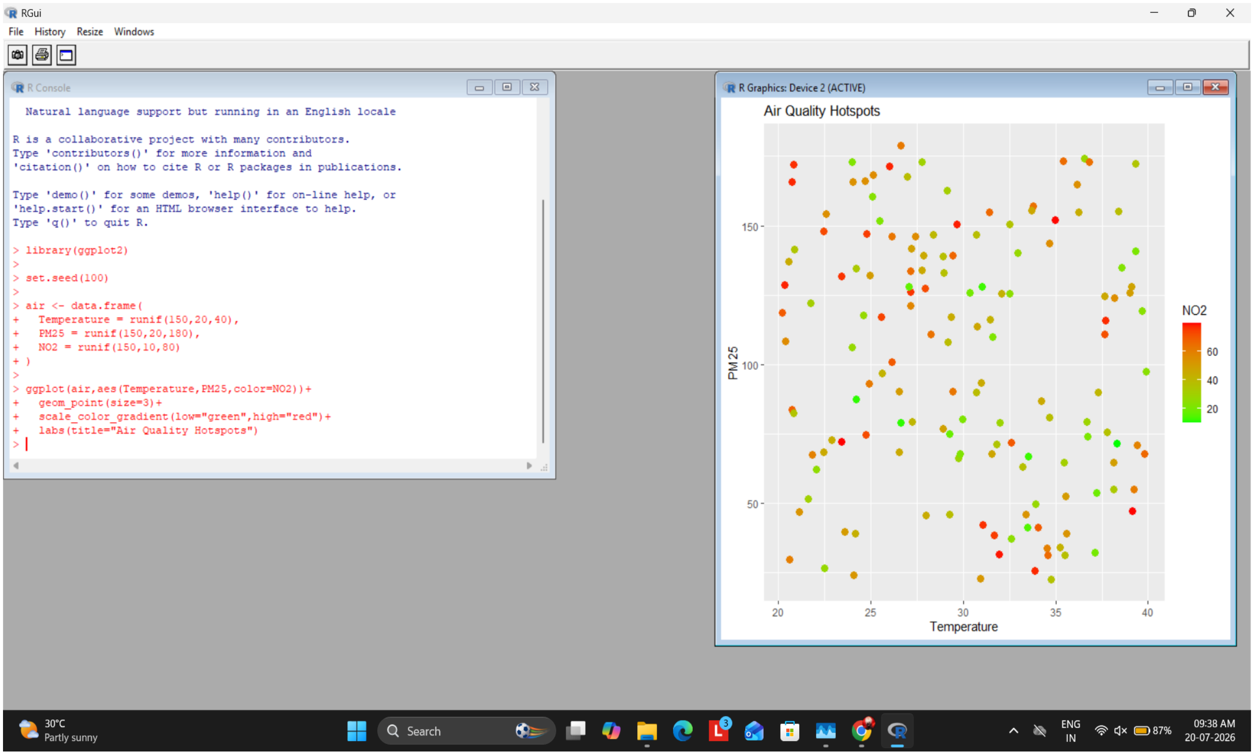
R Programs with Output Screenshots

Question 1

```
library(ggplot2)
set.seed(100)
air<-data.frame(Temperature=runif(150,20,40),PM25=runif(150,20,180),NO2=runif(150,10,80))
ggplot(air,aes(Temperature,PM25,color=NO2))+geom_point(size=3)+scale_color_gradient(low='green',high=
```

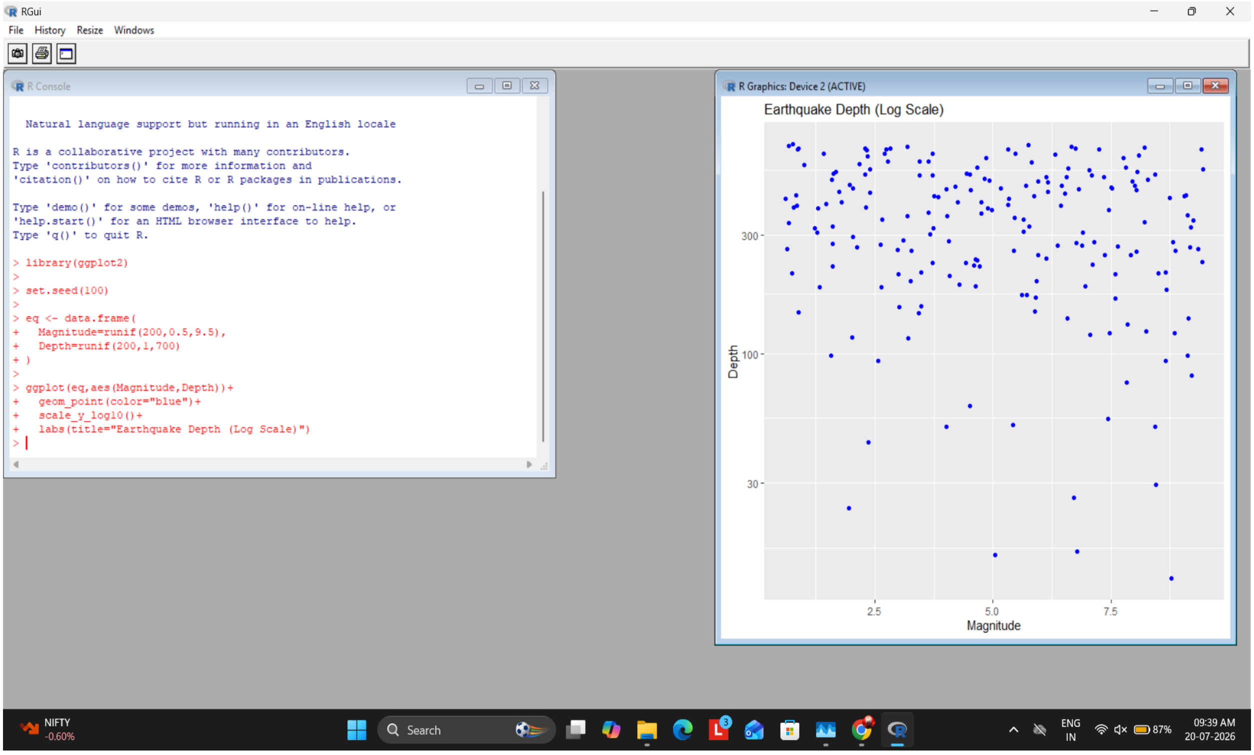
Output Screenshot:



Question 2

```
library(ggplot2)
set.seed(100)
eq<-data.frame(Magnitude=runif(200,0.5,9.5),Depth=runif(200,1,700))
ggplot(eq,aes(Magnitude,Depth))+geom_point(color='blue')+scale_y_log10()+labs(title='Earthquake Dept
```

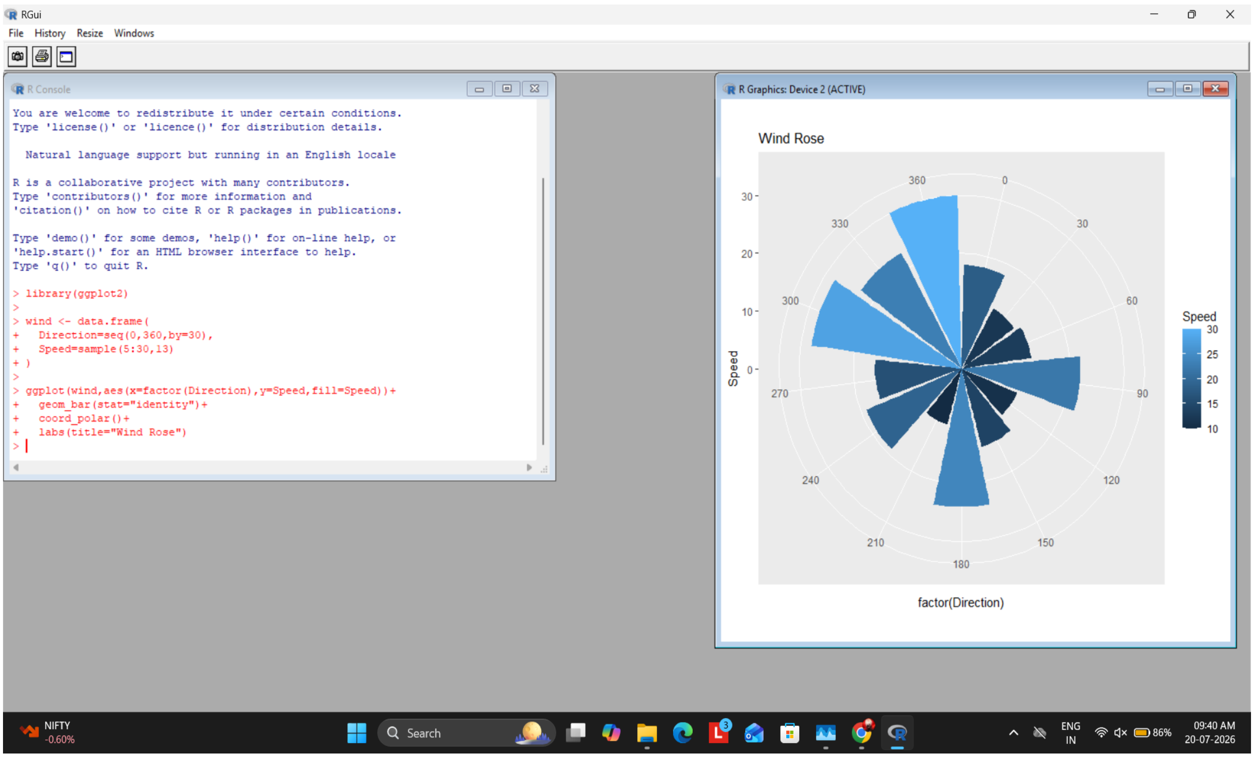
Output Screenshot:



Question 3

```
library(ggplot2)
wind<-data.frame(Direction=seq(0,360,30),Speed=sample(5:30,13))
ggplot(wind,aes(factor(Direction),Speed,fill=Speed))+geom_bar(stat='identity')+coord_polar()+labs(ti
```

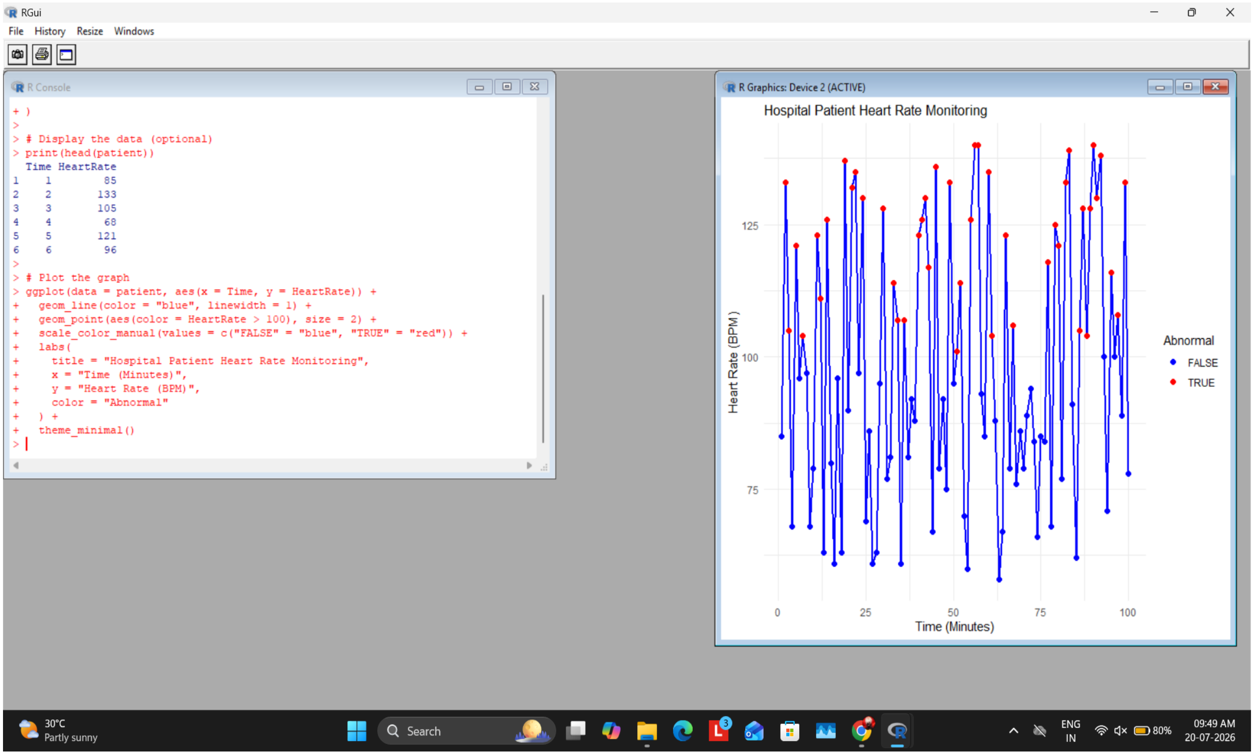
Output Screenshot:



Question 4

```
library(ggplot2)
set.seed(123)
patient<-data.frame(Time=1:100,HeartRate=sample(55:140,100,replace=TRUE))
ggplot(patient,aes(Time,HeartRate))+geom_line(color='blue')+geom_point(aes(color=HeartRate>100))+sca
```

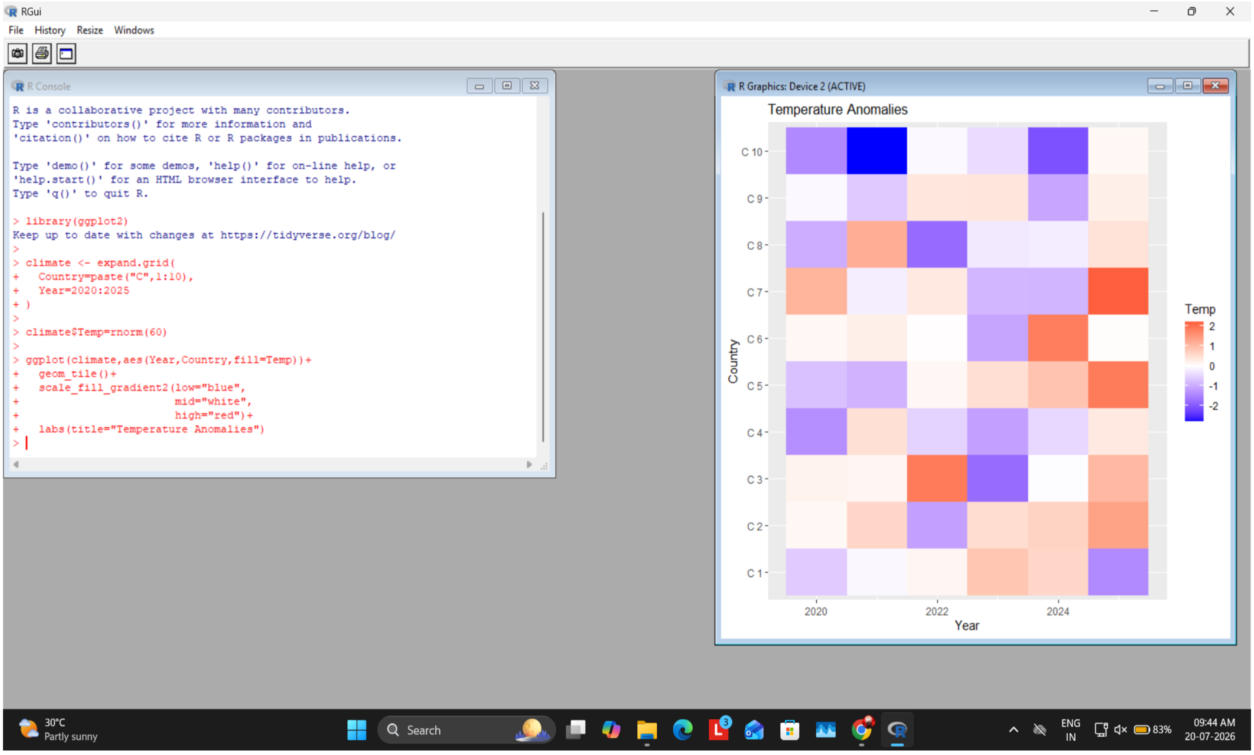
Output Screenshot:



Question 5

```
library(ggplot2)
climate<-expand.grid(Country=paste('C',1:10),Year=2020:2025)
climate$Temp=rnorm(60)
ggplot(climate,aes(Year,Country,fill=Temp))+geom_tile()+scale_fill_gradient2(low='blue',mid='white',
```

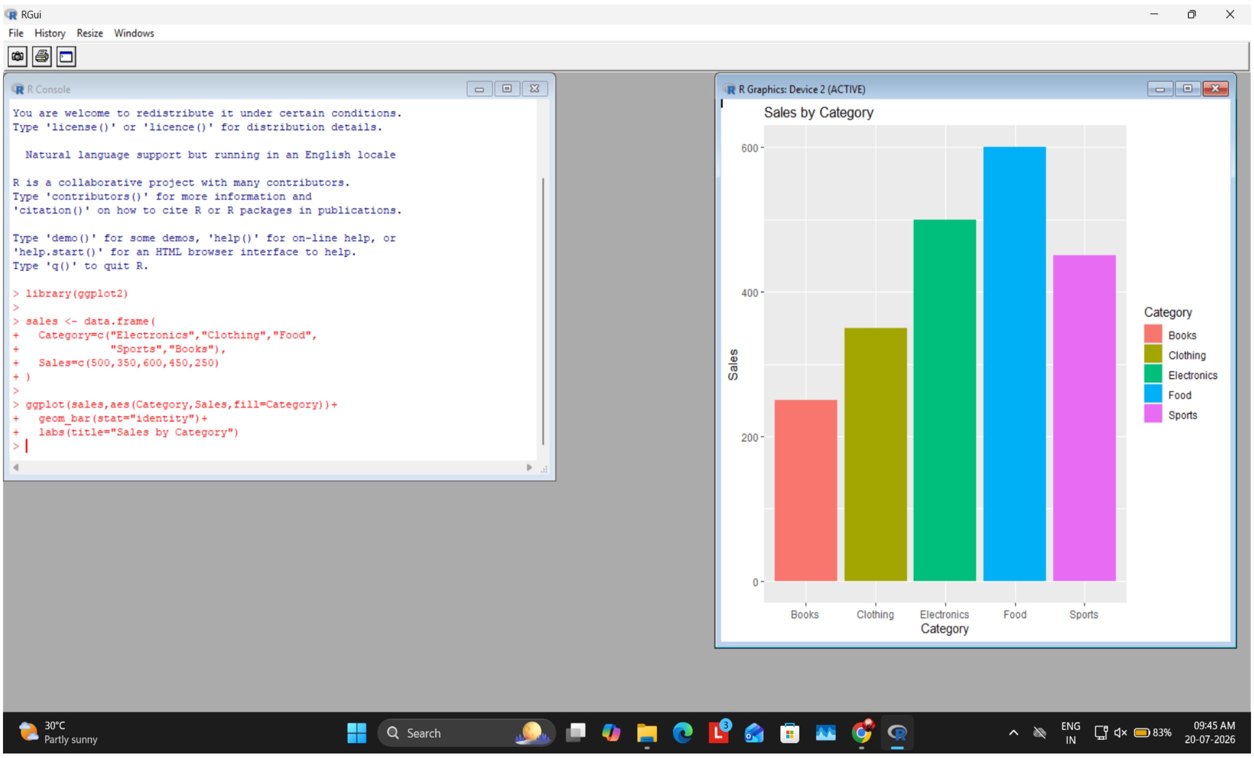
Output Screenshot:



Question 6

```
library(ggplot2)
sales<-data.frame(Category=c('Electronics','Clothing','Food','Sports','Books'),Sales=c(500,350,600,4
ggplot(sales,aes(Category,Sales,fill=Category))+geom_bar(stat='identity')
```

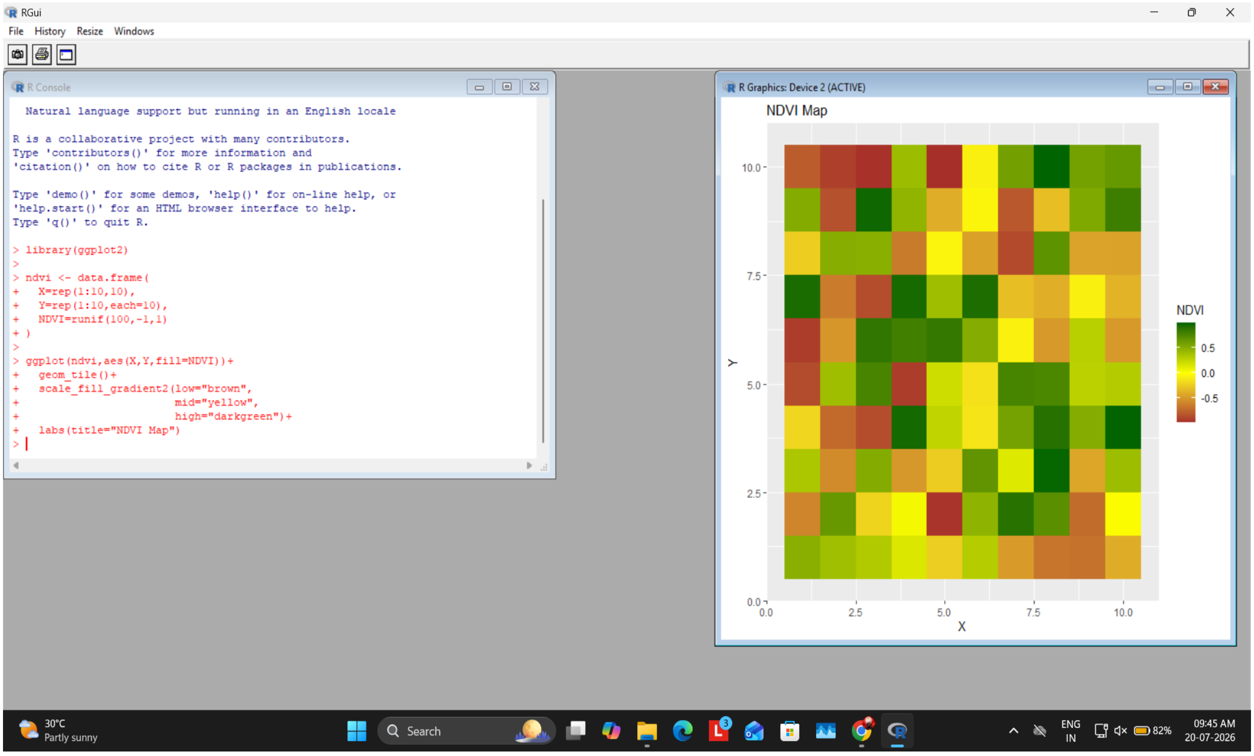
Output Screenshot:



Question 7

```
library(ggplot2)
ndvi<-data.frame(X=rep(1:10,10),Y=rep(1:10,each=10),NDVI=runif(100,-1,1))
ggplot(ndvi,aes(X,Y,fill=NDVI))+geom_tile()+scale_fill_gradient2(low='brown',mid='yellow',high='dark
```

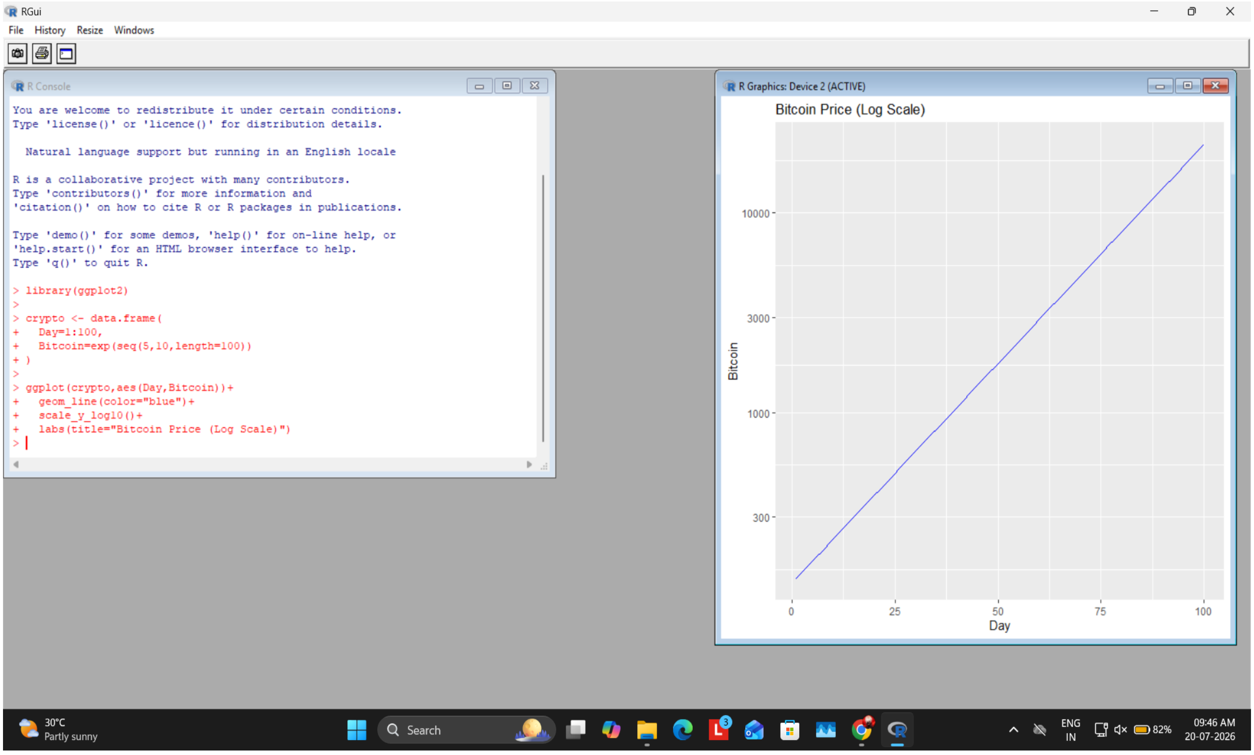
Output Screenshot:



Question 8

```
library(ggplot2)
crypto<-data.frame(Day=1:100,Bitcoin=exp(seq(5,10,length=100)))
ggplot(crypto,aes(Day,Bitcoin))+geom_line(color='blue')+scale_y_log10()
```

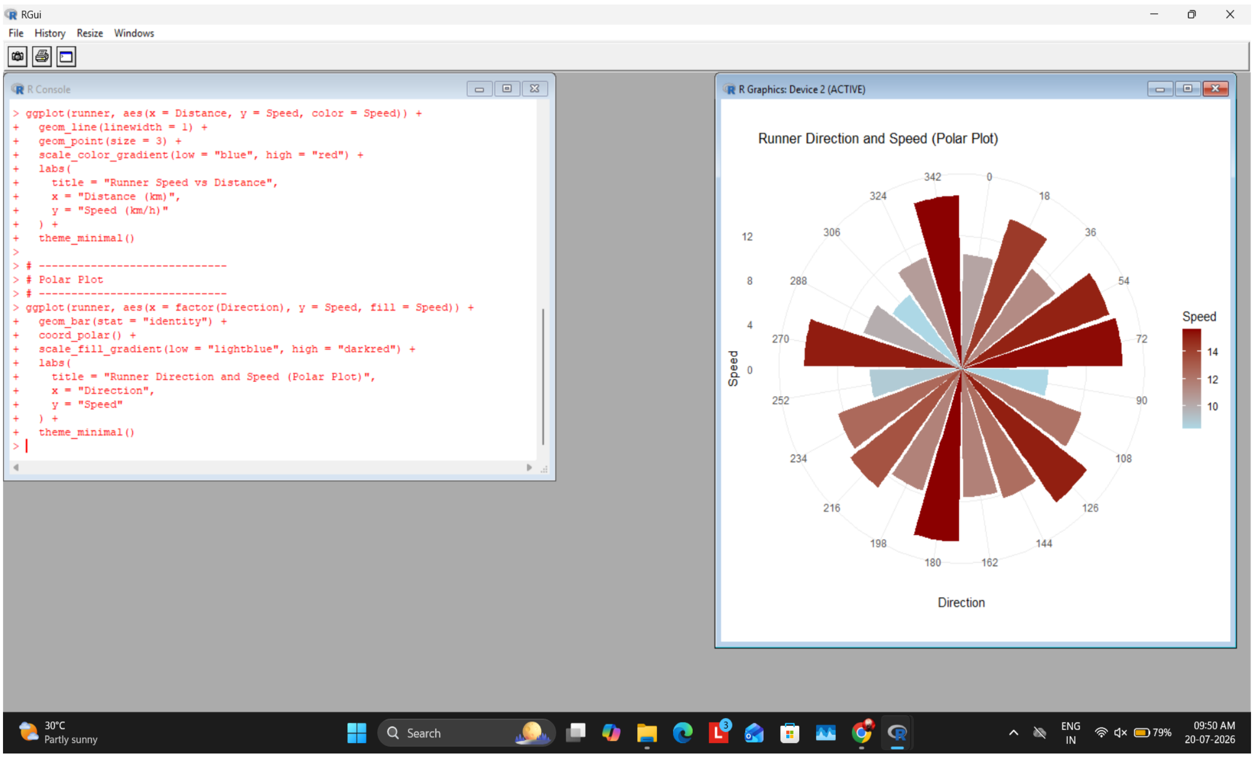
Output Screenshot:



Question 9

```
library(ggplot2)
runner<-data.frame(Distance=1:20,Speed=runif(20,8,16),Direction=seq(0,342,length.out=20))
# Cartesian and Polar plots
```

Output Screenshot:



Question 10

```
library(ggplot2)
vaccine<-data.frame(District=paste('D',1:10),Rate=sample(40:100,10))
ggplot(vaccine,aes(District,Rate,fill=Rate))+geom_bar(stat='identity')+scale_fill_gradient(low='yell
```

Output Screenshot: